Course Code	21AIE541T	Course Name	MULTIMODAL MACHINE LEARNING	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
							3	0	0	3

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Computational Intelligence*	Data Book / Codes / Standards	*Nil*		

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*provide the basic understanding of multimodal data and its importance in various fields*
CLR-2:	*various representations used in multimodal machine learning using different models*
CLR-3:	*understand the details about the translation and mapping algorithms of multimodal data*
CLR-4:	*create interest to develop a project using various applications of multimodal machine learning framework*
CLR-5:	*importance of multimodal deep learning and behavior generations functions*

		Program Outcomes (PO)												Program Specific Outcomes		
		1	2	3	4	5	6	7	8	9	10	11	12			
Course Outcomes (CO):	***At the end of this course, learners will be able to:***	Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	*outline the critical elements of multimodal data and models*	*3*	*3*	*3*	-	-	-	-	-	-	-	-	-	*1*	-	-
CO-2:	*illustrate different kinds of unimodal and multimodal representations*	*3*	*2*	*2*	-	-	-	-	-	-	-	-	-	-	*2*	-
CO-3:	*demonstrate multimodal translation and mapping*	*3*	*2*	*2*	-	-	-	-	-	-	-	-	-	*1*	-	-
CO-4:	*classify machine learning techniques and frameworks of multimodal applications in real time scenario*	*3*	*2*	*2*	-	-	-	-	-	-	-	-	-	-	-	-
CO-5:	*analyze various multimodal fusion and behavior generation for multimodal applications*	*3*	*2*	*2*	-	-	-	-	-	-	-	-	-	*1*	-	*2*

Unit-1 - Introduction ***9 Hour***

Introduction – Multimodal, Basic Concepts - Linear models - Score and loss functions – regularization, Neural networks - Activation functions - multi-layer perceptron, Optimization - Stochastic gradient descent – back propagation

Unit-2 - Unimodal and Multimodal Representations ***9 Hour***

Language representations - Distributional hypothesis and word embedding, Visual representations - Convolutional neural networks, Acoustic representations - Spectrograms – Auto encoders, Multimodal representations - Joint representations - Visual semantic spaces - multimodal auto encoder, Orthogonal joint representations - Component analysis, Parallel multimodal representations - Similarity metrics, canonical correlation analysis

Unit-3 - Multimodal Translation and Mapping ***9 Hour***

Language models – Unigrams – bigrams - skip-grams - skip-thought, Unimodal sequence modelling - Recurrent neural networks, LSTMs, Optimization - Back propagation through time, Multimodal translation and mapping - Encoder-decoder models - Machine translation - Image captioning, Generative vs retrieval approaches - Viseme generation - visual puppetry, Modality alignment - Latent alignment approaches - Attention models - multi-instance learning, Explicit alignment - Dynamic time warping

Unit-4 - Multimodal Applications ***9 Hour***

Multimodal fusion and co-learning - Model free approaches - Early and late fusion - hybrid models, Kernel-based fusion - Multiple kernel learning, Multimodal graphical models - Factorial HMM, Multi-view Hidden CRF, Case studies - Automatic Face Recognition - Video Segmentation and Keyframe Extraction - Gesture Recognition - Biometric-based System

Unit-5 - Deep Learning for Multimodal ***9 Hour***

Deep Learning for multimodal data fusion – Basics of multimodal deep learning – Multimodal image-to-image translation networks – Multimodal encoder decoder networks, Multimodal applications - Image captioning - Video description - AVSR, Core technical challenges - Representation learning – translation – alignment - fusion and co-learning

Learning Resources	1. *Multimodal Scene Understanding: Algorithms, Applications and Deep Learning, Michael Ying Yang, Bodo Rosenhahn, Vittorio Murino, Academic Press, Elsevier, 2019, ISBN:978-0-12-817358-9 (Unit V)* 2. *Representation Learning: A Review and New Perspectives. Yoshua Bengio, Aaron Courville, and Pascal Vincent* 3. *Visualizing and understanding recurrent networks. Andrej Karpathy, Justin Johnson, Li Fei-Fei, 2015*	4. *Unifying Visual-Semantic Embeddings with Multimodal Neural Language Models. Ryan Kiros, Ruslan Salakhutdinov, and Richard S. Zemel; TACL 2015* 5. *Multi-View Latent Variable Discriminative Models for Action Recognition. Yale Song, Louis-Philippe Morency, Randall Davis, CVPR 2012* 6. *M. Gori, "Machine Learning: A Constraint-Based Approach", 2017, Morgan Kauffman, ISBN: 978-0081006597* 7. *F. Camastra, A. Vinciarelli, "Machine Learning for Audio, Image and Video Analysis: Theory and Applications", 2nd Edition, 2016, Springer Verlag, ISBN: 978-1447168409*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	20%	-	20%	-
Level 2	Understand	20%	-	20%	-	20%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	30%	-	30%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers

Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. *Mr.Sanjay Hotwani, Senior Manager – Data Science, Products &Technology, PwC US*	1. *Dr. Tulasi Prasad Sariki, Associate Professor, VIT, Chennai, tulasiprasad.sariki@vit.ac.in*	1. *Dr.T.Subha, SRMIST*